

Statistics

1) Statistics:

Statistics is the science of collecting, organizing, analyzing and interpreting data to make decisions or draw conclusions.

i) Descriptive Statistics

Descriptive statistics summarizes and presents data using tables, graphs and numerical measures.

Ex: The average score of a class is 82%.

ii) Inferential Statistics

Inferential statistics uses sample data to make conclusions or predictions about a larger population.

Ex: Polling 1,000 voters to estimate how all voters in a country will vote.

2) Population:

The population is the entire group of individuals or items of interest in a study.

Ex: All students at a university.

3) Sample:

A sample is a subset of the population selected for study.

Ex: 200 students chosen from the university.

4) Variable:

A variable is a characteristic that can take different values for different individuals.

Ex: Height, age, or eye color.

5) Data:

Data are the observed values collected for variables.

- **Datum** (singular): One observation.
Ex: One student's age is 20.
- **Data** (plural): Multiple observations.
Ex: Ages of 200 students.

6) Experiment:

An experiment is a study in which researchers impose a treatment or condition and observe its effects.

Ex: Testing whether a new fertilizer increases plant growth.

7) Parameter:

A parameter is a numerical measure that describes a population.

Ex: The average age of all students at a university.

8) Statistic:

A statistic is a numerical measure calculated from a sample.

Ex: The average age of the 200 sampled students.

Types of Data:

1. Categorical Data:

Categorical data consists of labels or categories rather than numerical measurements.

Ex: Gender, blood type, favorite color, car brand.

2. Numerical Data:

Numerical data consist of numbers that represent counts or measurements.

a) Discrete Data

Discrete data are countable values, usually whole numbers.

Ex: Number of children in a family, number of cars owned, number of defects in a product.

b) Continuous Data

Continuous data can take any value within a range and are usually obtained by measurement.

Examples: Height (170.5 cm), weight (68.2 kg), temperature (22.7°C), time (3.45 seconds).

Levels of Measurement:

i) Qualitative:

A variable that categorizes or describes a population element.

ii) Quantitative:

A variable that quantifies a population element.

➔ Nominal, Ordinal

1. Nominal Data:

Nominal data consists of categories with no natural order or ranking.

Ex: Nationality, blood type.

2. Ordinal Data:

Ordinal data consists of categories that can be ranked, but the differences between ranks are not necessarily equal.

Ex: Class rank, education level

3. Interval Data:

Interval data are numerical data with equal intervals between values, but there is no true zero.

Ex: Temperature in °C or °F

4. Ratio Data:

Ratio data are numerical data with equal intervals and a true zero point.

Ex: Height, age, income

Measures of Central Tendency:

1) Mean:

The mean (average) is the sum of all values divided by the number of values.

2) Median:

The median is the middle value when data are arranged in order.

Ex: 3, 5, 7, 9, 11

Median = 7

3) Mode

The mode is the value that occurs most frequently.

Ex: 2, 3, 3, 4, 5

Mode = 3

Measures of Dispersion:

1) Range:

The range is the difference between the largest and smallest values.

2) Variance:

The variance measures the average squared deviation of data values from the mean.

Ex: If scores are very spread out, the variance is large; if scores are clustered near the mean, the variance is small.

3) Standard Deviation

The standard deviation is the square root of the variance. It measures the typical distance of data values from the mean.

Ex: If the mean test score is 80 and the standard deviation is 5, most scores tend to be within about 5 points of 80.

1) Symmetrical Distribution:

A symmetrical distribution has data evenly distributed around the center. The left and right sides are approximately mirror images.

Mean = Median = Mode

2) Asymmetrical Distribution

An asymmetrical distribution is not balanced around its center. One side extends farther than the other. The distribution is skewed either left or right.

3) Right-Skewed Distribution:

A right-skewed distribution has a long tail extending to the right (toward larger values).

4) Left-Skewed Distribution:

A left-skewed distribution has a long tail extending to the left (toward smaller values).

5) Bell Curve:

A bell curve is a perfectly symmetrical, bell-shaped distribution known as the normal distribution.

Distributions:

1. Normal Distribution:

The normal distribution is a continuous, bell-shaped, symmetric distribution where most values cluster around the mean.

2. Bernoulli Distribution:

A Bernoulli distribution describes a single experiment with only two possible outcomes: success or failure.

Ex: Pass/fail

3. Binomial Distribution:

A binomial distribution describes the number of successes in a fixed number of independent Bernoulli trials.

Ex: Number of heads in 10 coin tosses.

4. Uniform Distribution:

A uniform distribution is a distribution in which all outcomes are equally likely.

Ex: Chances of getting 1 when a dice is rolled. (1/6)

5. Poisson Distribution:

A Poisson distribution models the number of times an event occurs in a fixed interval of time, area, distance or volume.

Ex: Number of phone calls received per hour.

1) Empirical Rule (68–95–99.7 Rule)

For a normal distribution:

- **68%** of observations lie within 1 standard deviation of the mean.
- **95%** lie within 2 standard deviations of the mean.
- **99.7%** lie within 3 standard deviations of the mean.

2) Standard Normal Distribution

The **standard normal distribution** is a special normal distribution with:

- Mean (μ) = 0
- Standard deviation (σ) = 1

A **z-score** indicates how many standard deviations a value is above or below the mean.

Ex: A z-score of +2 means a value is 2 standard deviations above the mean.

3) Standardization

Standardization is the process of converting a normal random variable X into a standard normal variable Z .

